so

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After the second jet, vessel pressure falls from to . The given conditions yield

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The kinetic energy gained by the expelled gas is

If vessel pressure reaches ambient pressure after jets, then

Jetting now stops. The total kinetic energy of all the expelled gas is

Using to , becomes

The bracket in term on the right of is an -term geometric series, so

Also, give

Taking natural logarithms of both sides of gives

Since , the approximation further simplifies to